AI's Water Consumption: An Ethical Evaluation Through Indigenous Ethics

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Framework Application: Indigenous Ethics and AI's Water Footprint

This report applies Indigenous Ethics, centered on interconnectedness, reciprocity, harmony, and relational accountability, to evaluate the water consumption of artificial intelligence systems. Unlike frameworks rooted in individual rights or aggregate utility, Indigenous ethics treats water not as a resource to be managed but as a relational entity to which communities bear obligations. AI's water use is not merely an efficiency problem; it is a violation of reciprocity.

The scale of that violation is significant. Training a single large language model consumes millions of liters of fresh water, and running GPT-3 for 10-50 queries expends approximately 500 milliliters per query batch (Ren, 2023). Projected global AI water withdrawal by 2027 is estimated at 4.2-6.6 billion cubic meters, exceeding the total annual water withdrawal of four to six Denmarks (Ren, 2023). Between 2021 and 2022, Google's onsite water consumption grew 20% and Microsoft's by 34%, driven substantially by AI expansion (Ren, 2023).

From an Indigenous ethical standpoint, these figures represent a rupture in the relational web. Whyte (2018) argues that Indigenous ecological ethics are structured around collective continuance, the capacity of communities, including non-human communities, to maintain conditions necessary for their flourishing. Water evaporated from drought-stressed regions for data center cooling is redistributed away from the ecosystems most dependent on it (Li et al., 2023), placing disproportionate burden on vulnerable communities. The near-total absence of water footprint data in AI model cards (Ren, 2023) compounds the harm: what cannot be named cannot be held accountable.

Framework Reflection: Strengths, Limitations, and Alternatives

Indigenous ethics brings two key strengths: an intergenerational temporal scope that evaluates decisions across decades of aquifer depletion rather than quarterly reporting cycles, and a focus on local community impact rather than global averages. These are precisely the dimensions absent from current AI governance discourse.

Its limitations are also real. Indigenous ethics is deliberately plural, with no single codified system applicable across all contexts (Whyte, 2018), making it difficult to operationalize in regulatory or corporate settings. It identifies the relational wrong but does not prescribe whether the remedy is closed-loop cooling, workload redistribution, or mandatory disclosure (Li et al., 2023). A hybrid approach pairing Indigenous relational accountability with consequentialist harm metrics may be the most actionable path forward, provided the calculus does not permit local harm in exchange for aggregate global benefit.

Personal Reflection: Responsibility as a Practitioner

Working in AI infrastructure, where deployment decisions are made with detailed attention to compute cost and energy efficiency but rarely to water consumption, I find the disclosure gap unsettling. The carbon footprint of AI has entered mainstream awareness. The water footprint has not, and that asymmetry is not accidental: water is locally consumed but globally invisible, making it a convenient omission.

I do feel an ethical responsibility. Ren (2023) notes that water consumption is calculable from the same server energy measurements used to report carbon, yet it is routinely omitted from model cards. Kanungo (n.d.) projects that AI training compute has doubled every 3.4 months since 2012, placing an escalating burden on shared freshwater systems. Practitioners have a duty to advocate for water footprint disclosure as a professional standard, treating water not as infrastructure overhead but as a shared inheritance.

AI Assistance Disclosure

Claude (Anthropic, claude-sonnet-4-6) was used to summarize and synthesize data from assigned readings and to support structural organization. All arguments, framework selection, and personal *reflections represent the author's own reasoning, reviewed and validated prior to submission per USD Code of Honor guidelines.*

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